SPATIAO -TEMPORAL ATTENTION MECHANISM ON MRI SLICE SEQUENCES FOR MULTIPLE SCLEROSIS DIAGNOSIS

PROJECT REPORT

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in partial fulfillment for the award of the degree

of

BACHELOR OF TECHNOLOGY

in

INFORMATION TECHNOLOGY



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NOVEMBER 2026

BONAFIDE CERTIFICATE

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ACKNOWLEDGEMENT

We wish to express our sincere thanks to almighty and the people who extended their help during the course of our work. We are greatly and profoundly thankful to our honorable Chairman, **Col. Prof. Vel. Shri Dr.R.Rangarajan B.E.(ELEC), B.E.(MECH), M.S.(AUTO)., D.Sc.,** Vice Chairman, to late **Dr.Sakunthala Rangarajan M.B.B.S.,** for facilitating us with this opportunity. We take this opportunity to extend our gratefulness to our respectable Chairperson & Managing Trustee **Smt. Rangarajan Mahalakshmi Kishore B.E., M.B.A.,** and our Assistant Director **Ms. Eskandhaa Kishore Kumar Vel** for her continuous encouragement.

We also record our sincere thanks to our honorable Principal, **Dr. Ramesh, Ph.D.** for his kind support to take up this project and complete it successfully. We would like to express our special thanks to our Head of the Department, **Dr. ANANTHA RAMAN G R, Ph.D.,** Department of Information Technology and our project coordinator **Dr. M. Rajesh Khanna, Ph.D.,** and supervisor **Mrs. J. MANJU, M.E** for their moral support by taking keen interest on our project work and guided us all along, till the completion of our project work.

Further, the acknowledgement would be incomplete if we would not mention a word of thanks to our most beloved parents for their continuous support and encouragement all the way through the course that has led us to pursue the degree and confidently complete the project work.

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

ABSTRACT

Multiple Sclerosis (MS) is a chronic autoimmune disease in which the immune system damages the myelin sheath that protects nerve fibers, resulting in neurological symptoms and potential disability over time. Magnetic Resonance Imaging (MRI) plays an essential role in MS diagnosis and monitoring by providing detailed images of the brain and identifying abnormalities. However, manually examining numerous MRI slices is time-consuming and challenging. To address these limitations, this project proposes a spatial-temporal deep learning framework with slice-sequence attention for automated MS diagnosis using multi-sequence brain MRI.

Existing approaches often use two-dimensional Convolutional Neural Networks (2D CNNs) to analyses individual slices independently, overlooking anatomical continuity between successive slices, while three-dimensional CNNs (3D CNNs) require greater computational resources and substantial labelled data. In addition, improper data splitting can cause data leakage when slices from the same patient appear in both training and testing sets, leading to overly optimistic results. The proposed framework combines T1, T2, and FLAIR MRI modalities into three-channel axial images and applies standardized preprocessing techniques, including skull stripping, bias field correction, intensity normalization, and resizing.

Ordered sequences of anatomically contiguous MRI slices are processed using EfficientNet-B0 for spatial feature extraction, a Bidirectional Long Short-Term Memory (BiLSTM) network for sequential feature learning, and slice-sequence attention to emphasize diagnostically relevant slices. Subject-level five-fold cross-validation is proposed to reduce cross-subject data leakage and provide a more reliable assessment of model generalization. Furthermore, Explainable Artificial Intelligence (XAI) techniques, including slice-level attention visualization and Gradient-weighted Class Activation Mapping (Grad-CAM), are incorporated to highlight MRI slices and image regions influencing the model's predictions, improving interpretability and supssporting the development of a more reliable automated MS diagnosis framework.

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LIST OF ABBREVIATIONS

ANN	Artificial Neural Network
ASPP	Atrous Spatial Pyramid Pooling
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
BiLSTM	Bidirectional Long Short-Term Memory
CAD	Computer-Aided Diagnosis
FLAIR	Fluid-Attenuated Inversion Recovery
GLCM	Gray-Level Co-occurrence Matrix
Grad-CAM	Gradient-weighted Class Activation Mapping
MRI	Magnetic Resonance Imaging
MS	Multiple Sclerosis
ROI	Region of Interest
SVM	Support Vector Machine
WHM	White Matter Hyperintensity
XAI	Explainable Artificial Intelligence

CHAPTER 1
INTRODUCTION

1.1 DEFINITION

Multiple Sclerosis (MS) is an autoimmune disease affecting the central nervous system and is characterized by demyelination. The disorder provokes various neurological symptoms depending on the site of involvement. MRI scan interpretation is frequently used in MS diagnosis because it provides cross-sectional images of the brain for lesion detection. However, manually analyzing the resulting images is cumbersome, especially since a single MRI includes dozens of images. A solution to this issue lies in implementing diagnostic algorithms capable of processing the data for faster and more accurate diagnosis.

To begin with, the information received from scans is multi-layered and requires expert analysis, which takes much time and effort. Furthermore, the visual inspection of MRI images is highly subjective and can be improved by applying a standardized automated procedure. In addition, the current workflow can lead to operator fatigue and inconsistent results due to human factors. Finally, implementing an automatic procedure will increase the efficiency and accuracy of MS diagnosis since it will reduce the possibility of human error. The complex information obtained from MRI demands for time-consuming interpretation, while the visual inspection is highly subjective, often resulting in operator fatigue. An automated analysis could standardize the procedure, reducing the possibility for human error and increasing the accuracy and efficiency of the MS diagnosis.

1.2 PROBLEM STATEMENT

A patient's brain MRI consists of many slices taken in succession, and many existing 2D Convolutional Neural Network (CNN) methods process each image independently, thus ignoring the important anatomical continuity and relations between consecutive slices. 3D CNN methods process a volume of MRI scans, and though they preserve spatial relationships in all three dimensions, they require far more memory and processing power, which can be a limiting factor. Moreover, some papers have overly optimistic results due to patient-level data leakage during cross-validation, where a slice of the same patient's data was in both the training and test sets by chance.

1.3 OBJECTIVES

The main goal of the project is to devise a deep learning method for analyzing multi-sequence magnetic resonance imaging (MRI), including T1, T2, and FLAIR scans. The proposed method will aim to utilize conventional preprocessing steps such as skull stripping, bias field correction, intensity normalization, and image resizing to enhance the quality of input data and ensure consistency across various sequences and subjects. Furthermore, the method will seek to organize MRI slices into consecutive sequences to maintain anatomical continuity and employ features extracted from individual MRI slices using an EfficientNet-B0 convolutional neural network and a BiLSTM-based slice-sequence encoder-decoder to capture associations between MRI slices and predict subject labels based on acquired features. A slice-sequence attention will be implemented to enable the model to identify the MRI slices that are the most informative for the downstream task. The resulting model will be evaluated using a leave-five-subjects-out cross-validation procedure to ensure that the train and test sets do not overlap and to prevent.

Lastly, Grad-CAM will be applied to visualize the ROIs that the model pays the most attention to, thus explaining the model's decisions and improving its interpretability.

1.4 SCOPE OF THE PROJECT

The scope of this project is to engineer an automated deep learning model for the diagnostic needs of Multiple Sclerosis from axial brain magnetic resonance imaging (MRI). The system is suggested as a research prototype and a computer-aided diagnostic framework that takes advantage of spatial-sequential learning and explainable artificial intelligence. The current proposal includes the development process, design, training, evaluation, visualization, and analysis of the diagnostic system. However, it is not intended to provide medical treatment or substitute professional diagnosis.

1.5 MOTIVATION AND SIGNIFICANCE

Multiple Sclerosis is a serious disease and affects 2.9 million people worldwide by 2023. This disorder is less common in India than in the Western world, but it is still significant given the country's large population. When a radiologist reads an MRI scan in a hospital, they move through slices one by one to get the overall picture of lesions. This project is inspired by the way medical professionals review MRIs in day-to-day practice. It is proposed to process a set of MRI images as a sequence of slices to build a complete picture of lesions for better diagnosis and well-being of the patients. The approach aims to exploit information within individual slices and the adjacent ones to learn the distribution of MS lesions effectively.

CHAPTER II
LITERATURE SURVEY

2.1 REVIEW OF EXISTING RESEARCH

Previously, automated MS (Multiple Sclerosis) detection systems were largely built upon conventional machine learning algorithms, where manual feature extraction was a time-consuming and error-prone task [2]. Recent studies have turned to utilizing deep learning approaches. 2D CNN-based methods have proven particularly useful for the task due to their relatively low resource requirements; however, since they process images sequentially, they lose valuable anatomical information about the connectivity between different slices. 3D CNN approaches learn features collectively from the entire MRI volume; however, they demand significant computational resources and are challenging to apply to smaller datasets [8]. Recent research in vision-related tasks has shown the effectiveness of attention mechanisms in focusing on disease-specific regions [9], and approaches based on Transformer encoder-decoder architectures have proven beneficial for medical imaging tasks [10].

2.1.1 Classical Machine Learning Approaches in Neuroimaging

Early Computer-Aided Diagnosis (CAD) systems relied on manual Region of Interest (ROI) selection, thresholding, and handcrafted features. Traditional approaches incorporated texture-based statistical features such as Gray-Level Co-occurrence Matrices (GLCM), Local Binary Patterns, and Wavelet Transforms to characterize structural variations in brain tissue. Regarding feature-engineered classifiers, combining handcrafted texture features with algorithms like Support Vector Machines (SVM), Random Forests, or Linear Discriminant Analysis can yield promising results in small-scale studies; however, their performance is often compromised by factors such as scanner heterogeneity, variations in image intensity, and operator-dependence

Variability [11]. In the realm of metaheuristic optimization, recent research suggests optimizing the feature space through evolutionary algorithms. Anam et al. (2026) utilized a Whale Optimization Algorithm to prune redundant information from deep learning features extracted via a VGG16 network, subsequently employing an Artificial Neural Network (ANN) classifier to achieve 85% accuracy on benchmark datasets [3]. Nevertheless, due to sequential model architecture, the process of encoding spatial information is decoupled from the classification task, resulting in the loss of high-order spatial correlations between consecutive MRI slices.

2.1.2 Convolutional Neural Networks (CNNs) in Brain MRI Analysis

Convolutional Neural Networks (CNNs) eliminate manual feature engineering by learning hierarchical spatial representations directly from raw voxel intensities [12].

2D CNN Architectures: 2D CNN backbones process individual slices, enabling rapid inference and computational efficiency [13]. Bp et al. (2024) implemented an EfficientNet-B1 backbone leveraging transfer learning on axial and sagittal FLAIR planes. This demonstrated that CNNs utilizing transfer learning can detect local demyelination patterns with a relatively small number of parameters [8]. However, analyzing slices individually ignores connectivity along the z-axis, thereby preventing the modelling of volumetric relationships between plaques.

3D CNN Formulations: Studies have employed 3D convolutions for both lesion segmentation and overall patient classification. Saeed et al. (2025) proposed a multi-path 3D CNN integrated with conditional random fields to segment 3D Multiple Sclerosis (MS) lesions of varying scales, which helps reduce false positives [16]. Nevertheless, because 3D convolutions involve an exponential increase in the number of parameters, they often lead to computational bottlenecks and a heightened risk of overfitting when working with small training datasets [15].

2.1.3 Encoder-Decoder Architectures and Spatial Context

Semantic segmentation models, especially U-Net variations, serve as the first-line method for delineating spatial lesion distributions.

Contextual Aggregation: The standard U-Net variants have difficulty handling diffuse plaque edges and thin juxtacortical lesions. Jain et al. (2026) proposed MS-DASPNet that employs a pre-trained VGG-16 encoder with an atrous spatial pyramid pooling (ASPP) bottleneck and dual-headed attention modules for the skip connections, allowing contextual information from multiple scales to be aggregated through the attention mechanism. It achieved a dice score of 0.8736 on MICCAI-2016 [6].

Volumetric Reconstruction Constraints: Individual MRI slices may not capture the complete structure and distribution of Multiple Sclerosis (MS) lesions, making it necessary to consider the contextual relationships between consecutive slices. The model must extract meaningful spatial features, including lesion shape, size, texture, and location, while preserving anatomical continuity through the correct ordering of MRI slices. Integrating T1-weighted, T2-weighted, and FLAIR sequences can provide complementary information for identifying disease-related patterns. Furthermore, variations in lesion appearance and distribution across patients present challenges for consistent prediction. Since the final output requires patient-level classification, features extracted from individual slices must be effectively aggregated to generate a reliable prediction for the entire MRI volume. A slice-sequence attention mechanism can help identify the most informative slices and reduce the influence of less relevant regions.

2.1.4 Transformer-Based Frameworks and Attention Mechanisms

To model long-range spatial dependencies without the localized inductive bias of standard convolutions, Vision Transformers (ViTs) and hybrid attention networks have emerged in medical imaging.

- **Diffusion and Channel Attention:** Anilkumar et al. (2026) introduced the Self-Channel Attention Diffusion Model (SCADM), combining self-channel attention with iterative diffusion steps to refine feature distributions and suppress scanner noise, achieving a 94.44% accuracy on clinical MRI splits [5].
- **Dual-Attention Transformers (MS-Trust Base Paper Analysis):** The base paper by Alzahrani (2025) presents MS-Trust, a transformer model featuring a Compact Convolutional Tokenizer (Tokenizer) coupled with causal-global dual attention and Squeeze-and-Excitation (SE) blocks [1]. The causal attention block preserves sequential ordering, while the global attention block captures broad structural relationships across image patches. Although MS-Trust demonstrated 90% to 92.85% accuracy in distinguishing MS, Myelitis, and Healthy controls, its architecture requires repetitive encoder passes and dense matrix multiplications, remaining computationally intensive and lacking explicit recurrent sequence tracking across whole anatomical slices [1].

2.1.5 Explainable Artificial Intelligence (XAI) in Clinical Neuroimaging

Deep neural networks operate as opaque systems, creating validation challenges for clinical adoption. Explainable AI (XAI) methodologies provide visual interpretations of network decisions.

Class Activation Mapping: Gradient-weighted Class Activation Mapping (Grad-CAM) computes the gradients of the target class score with respect to feature maps in the final convolutional layer, producing coarse localization heatmaps of influential regions [7].

Artifact Detection and Model Trust: Bibi et al. (2024) highlighted that initial Grad-CAM heatmaps exposed network bias toward background scanner artifacts and skull tissue, allowing researchers to correct the preprocessing pipeline via targeted cropping and re-training [7]. Similarly, Zhang et al. (2021) demonstrated that Grad-CAM and Grad-CAM++ reliably map demyelinating lesions to validated radiological phenotypes, bridging the trust gap between automated models and clinical decision makers.[7]

2.1.6 Summary of Literature Analysis

The literature establishes that 2D CNNs offer computational efficiency but discard slice-to-slice continuity, whereas 3D CNNs and pure vision transformers capture volume context at the expense of high computational demand and potential overfitting. While attention and transformer models (such as MS-Trust) effectively weight salient features, they do not incorporate ordered recurrent modeling across full contiguous slices. This observation provides the foundation for the proposed framework: coupling lightweight 2D spatial feature extractors with bidirectional recurrent sequence learning, slice-level self-attention, and Grad-CAM visualization.

2.2 CHALLENGES IDENTIFIED

2.2.1 Inter-Slice Spatial Discontinuity in 2D Modelling

Treating volumetric MRI scans as decoupled 2D images fails to represent the physical reality of human neuroanatomy. Multiple sclerosis plaques are three-dimensional structures that span several adjacent axial slices. Independent 2D evaluations cause the network to treat consecutive sections of the same lesion as disjoint features, losing vital progression metrics along the inferior-superior axis.

2.2.2 High Computational Complexity of 3D Volumetric Convolutions

Direct 3D volumetric modelling using full 3D kernels escalates memory allocation, floating-point operations (FLOPs), and gradient storage requirements. These architectures demand enterprise GPU clusters and substantial training data, making them impractical for standard hospital infrastructure and prone to overfitting on small medical cohorts.

2.2.3 Data Leakage and Over-Optimistic Generalization

A widespread issue in published literature is slice-level data partitioning. When random train-test splitting is conducted at the slice level rather than the patient level, slices from the same individual appear in both training and test partitions. Because consecutive slices share identical anatomical morphology, scanner noise profiles, and brain contours, models memorize patient identity rather than disease pathology, producing over-optimistic accuracy metrics that fail on external cohorts.

2.2.4 Difficulty in Differentiating Benign versus Pathological Hyperintensities

On T2-FLAIR imaging, not all hyperintense signals denote active demyelination. As established by Bawil et al. (2026) in the MS3SEG dataset, non-pathological age-related white matter hyperintensities and cerebrospinal

Fluid (CSF) partial volume effects around ventricular borders frequently mimic MS plaques, leading automated CNNs to produce high false-positive rates unless explicit structural context is provided.

2.2.5 Unbalanced Clinical Datasets and Anisotropic Acquisitions

Medical image repositories are constrained by small patient cohorts and high-class imbalance, where background tissue comprises over 99% of total voxels. Furthermore, routine clinical scans frequently utilize 2D thick-slice protocols (e.g., 5 mm slice thickness with inter-slice gaps) rather than isotropic 3D acquisitions, introducing partial volume artifacts that degrade spatial boundary detection.

2.2.6 Lack of Clinical Interpretability in Deep Black-Box Models

End-to-end classification decisions without visual explanations present significant barriers to clinical integration. Without verifying which slices and specific anatomical regions influenced a diagnostic prediction, clinicians cannot determine whether a network's output is driven by genuine demyelinating lesions or scanner-specific background artifacts.

2.3 OBJECTIVES OF PROPOSED WORK

To address the limitations of 2D slice independence, 3D computational bottlenecks, data leakage, and opaque decision boundaries, the proposed framework is divided into the following specific engineering objectives:

2.3.1 Standardized Multi-Sequence MRI Preprocessing and Alignment

Develop a standardized preprocessing pipeline for multi-sequence brain MRIs (T1, T2, FLAIR). This includes rigid-body registration to map sequences into a unified space, brain extraction (skull stripping) to eliminate non-brain artifacts, N4 bias field correction to resolve scanner inhomogeneities, and intensity normalization to standardize voxel distributions.

2.3.2 Multi-Sequence Stacking and Ordered Axial Slice Formation

Construct three-channel composite images by stacking aligned T1, T2, and FLAIR slices along the channel dimension, leveraging complementary tissue contrasts (structural detail from T1, fluid suppression and lesion conspicuity from FLAIR). Arrange these slices into an anatomically ordered axial sequence per subject to preserve inferior-to-superior geometric continuity.

2.3.3 Lightweight Spatial Feature Extraction via EfficientNet-B0

Integrate a pre-trained EfficientNet-B0 convolutional backbone to extract compact, high-level spatial representations from each composite slice. EfficientNet-B0 utilizes compound scaling (balancing network depth, width, and resolution via mobile inverted bottleneck convolutions) to generate rich feature embeddings while maintaining minimal parameter overhead compared to heavier architectures.

2.3.4 Ordered Slice-Sequence Modelling Using Bidirectional LSTM

Incorporate a Bidirectional Long Short-Term Memory (BiLSTM) network to process the sequence of spatial feature vectors extracted from the contiguous slices. By analysing feature changes in both forward (inferior-to-superior) and backward directions, the BiLSTM captures inter-slice dependencies and models how lesion boundaries transition across the z-axis.

2.3.5 Implementation of Slice-Sequence Self-Attention

Design a slice-level self-attention mechanism over the BiLSTM hidden states to dynamically compute relevance weights for each anatomical slice. Because MS plaques are localized and only present in a subset of the full scan, the attention module highlights diagnostically critical slices containing active lesions while down-weighting uninformative or healthy slices prior to classification.

2.3.6 Patient-Level Classification and Decision Logic

Construct a dense classification head that aggregates the attended sequence representation to predict patient-level Multiple Sclerosis status. The classification network outputs calibrated probability scores, providing binary diagnosis while preserving end-to-end differentiability.

2.3.7 Transparent Explainability via Slice Attention and Grad-CAM

Integrate Explainable AI (XAI) mechanisms to provide transparent dual-stage interpretability. Use slice-level attention weights to identify which axial slices contributed most significantly to the final classification and apply Gradient-weighted Class Activation Mapping (Grad-CAM) to the final convolutional layers of the EfficientNet-B0 backbone for those slices, generating visual heatmaps that overlay precisely on detected lesions.

2.3.8 Strict Subject-Level Five-Fold Cross-Validation

Establish an evaluation protocol using strict subject-level 5-fold cross-validation. Group all slices and sequences belonging to a single subject into a single partition, ensuring that no patient's data is shared between training and validation sets, eliminating data leakage and providing an accurate assessment of generalizability. All preprocessing parameters, feature selection procedures, and model training operations that learn from data must be fitted using only the training portion of each fold. Hyperparameter tuning and threshold selection should be performed without using the corresponding validation fold as an independent test set. Model performance should be evaluated using appropriate metrics, such as accuracy, precision, recall, F1-score, and the area under the Receiver Operating Characteristic curve (ROC-AUC). The results from all five folds should be summarized using the mean and standard deviation to measure performance consistency and variability.

CHAPTER III

REQUIREMENT ANALYSIS

3.1 FUNCTIONAL REQUIREMENTS

- The system shall accept T1-weighted, T2-weighted, and FLAIR MRI as input.
- The system shall perform skull stripping, bias field correction, and intensity normalization as preprocessing steps.
- The system shall stack the images and extract ordered sequences of axial slices.
- The system shall perform MS-related classification based on the image features identified.
- The system visualizes the classification results using Grad-CAM.

3.2 NON-FUNCTIONAL REQUIREMENTS

- Performance: The model must be computationally efficient enough to run on standard research GPU hardware.
- Reliability: The evaluation must use strict subject-level cross-validation to guarantee realistic performance metrics.
- Usability: The final predictions should be easily interpretable through visual heatmaps.
- Frontend Responsiveness: The web dashboard must provide a responsive interface that works across desktops, laptops, and tablets.
- User Interface Consistency: The dashboard must maintain consistent layouts, navigation, typography, colours, and visual components across all pages.

3.3 HARDWARE REQUIREMENTS

- Processor: Modern Multi-core CPU.
- Memory: Minimum 16 GB RAM.
- GPU: NVIDIA GPU with sufficient VRAM (e.g., RTX series) for deep learning training and inference.

3.4 SOFTWARE REQUIREMENTS

- Operating System: Windows / Linux.
- Programming Language: Python 3.x.
- Frameworks: TensorFlow / Keras or PyTorch for deep learning.
- Libraries: OpenCV, NumPy, Scikit-learn, Matplotlib.
- Modeling Tool: StarUML for system architecture and design diagrams.

3.5 EXISTING SYSTEM

The existing systems are primarily reliant on either isolated 2D CNNs or heavier 3D CNNs. The 2D CNNs process each MRI slice in isolation and ignore anatomical continuity between successive slices. 3D CNNs require heavy memory consumption and processing power for analysing the entire volumetric data. In addition, many systems tend to have patient-level data leakage while training the models and lack sufficient visual explanation (XAI) to aid clinicians in understanding the system's prediction.

3.6 PROPOSED SYSTEM

The designed system is the spatial-temporal deep learning framework with slice-sequence attention for multiple sclerosis (MS) diagnosis. By considering multi-sequence MRI scans as a series of slices in a specific order, the system simulates the reading process of a radiologist. The framework uses an EfficientNet-B0 module to extract spatial features from each slice of the brain, a bidirectional long short-term memory (BiLSTM) module to encode the spatial information of all slices in the sequence, and an attention module to focus on the most diagnostic slices in the sequence.

3.7 FEASIBILITY STUDY

Technical Feasibility: Using the EfficientNet-B0 assures the researcher that the process of feature extraction will be less resource-heavy than the use of a standard 3D CNN. The implementation of a BiLSTM for sequence learning has proven to be a technically viable solution.

Operational Feasibility: The production of Grad-CAM heatmaps in conjunction with predictive results assuages the concern that the solution will provide clinically relevant insights with a satisfactory degree of explainability.

Economic Feasibility: The project's reliance on open-source software and the use of readily available data ensures the economic viability of the solution.

3.8 Proposed System Architecture

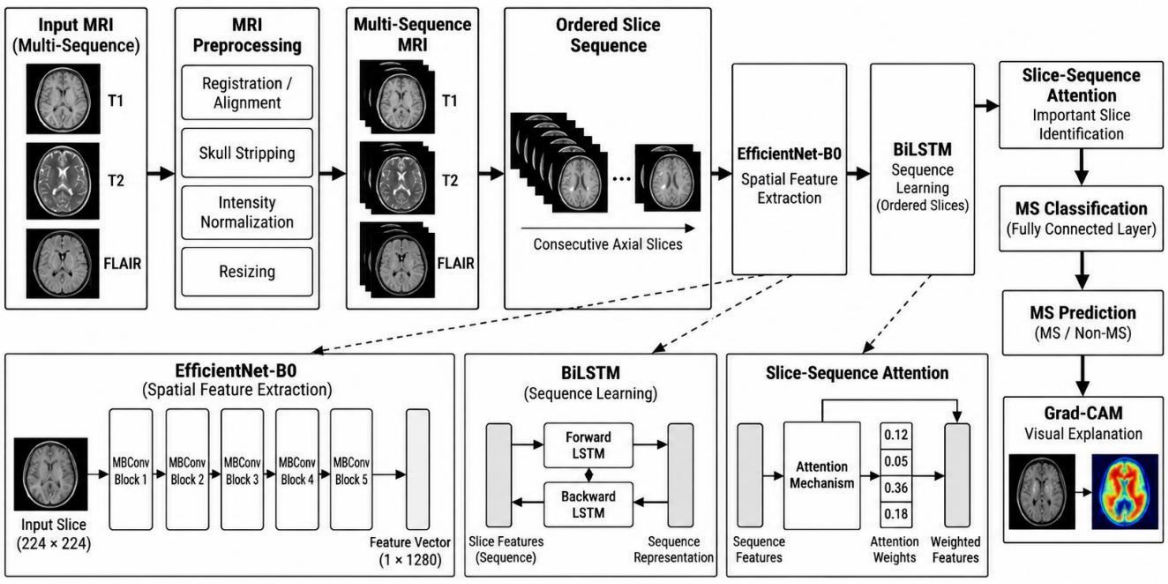


Figure: Proposed System Architecture for Multiple Sclerosis Detection

Figure 3.1 Proposed System Architecture for Multiple Sclerosis Detection

The proposed architecture follows a clear pipeline:

1. **Input MRI:** Multi-sequence T1, T2, and FLAIR images are collected.
2. **MRI Preprocessing:** The images undergo registration, skull stripping, intensity normalization, and resizing.
3. **Multi-Sequence MRI:** The modalities are combined into three-channel composite images.
4. **Ordered Slice Sequence:** Consecutive axial slices are extracted as an ordered sequence.
5. **EfficientNet-B0 Spatial Feature Extraction:** Each slice in the sequence is passed through EfficientNet-B0 to extract a spatial feature vector.
6. **BiLSTM Sequence Learning:** The feature vectors are processed by a BiLSTM to learn the forward and backward sequential relationships.
7. **Slice-Sequence Attention:** An attention mechanism identifies and weights the most important slices in the sequence.
8. **MS Classification:** A fully connected layer processes the attended features to output the final MS prediction.
9. **Grad-CAM:** Visual explanations are generated to highlight the regions influencing the classification.

3.9 UML DIAGRAMS

3.9.1 Use Case Diagram

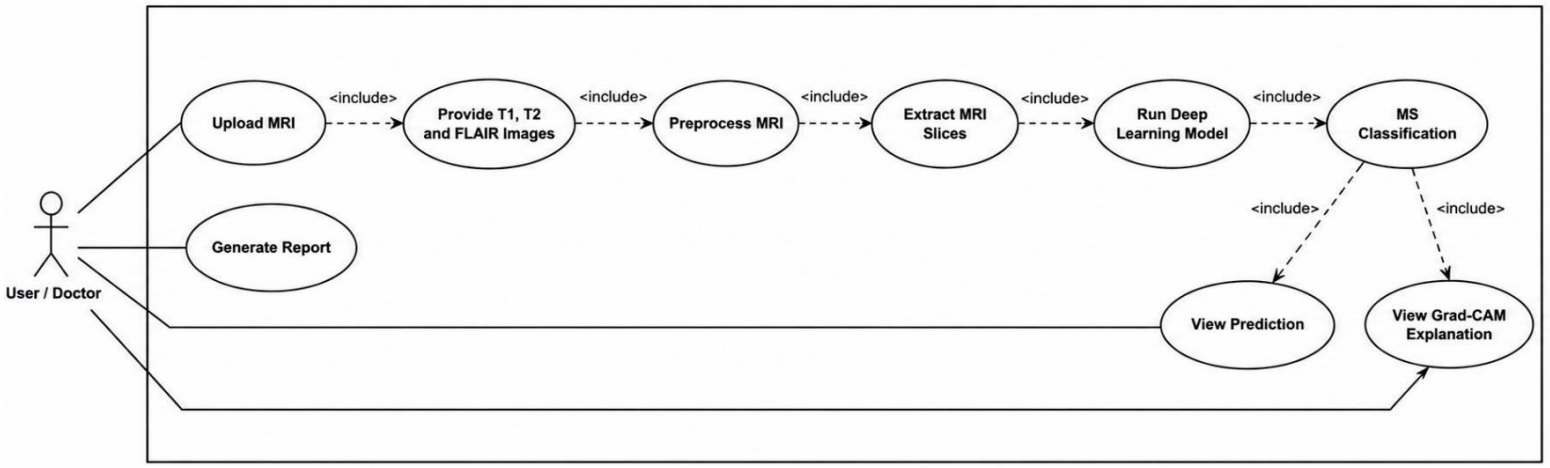


Figure 3.2 Use Case Diagram

Figure 3.2 illustrates the Use Case diagram shows what the end-to-end workflow of the Multiple Sclerosis Diagnosis System looks like from the perspective of a User / Doctor, including how features help support accurate detection of MS and provide an explanation. A User / Doctor begins by initiating several possible primary actions: uploading an MRI scan, viewing predictions, viewing Grad-CAM explanations, or generating a diagnostic report. After an MRI is uploaded, the system will automatically initiate a process of included processes to deliver T1, T2, and FLAIR images, preprocess the MRI data, extract the MRI slices, and run the deep learning model to final MS classification. To ensure the clinical interpretation is straightforward, MS classification process will directly allow you to view the final prediction and view Grad-CAM explanations to understand what the model has learned. It is also possible for the doctor to independently generate a comprehensive report based on the findings from the system. Overall, the diagram illustrates how the proposed system integrates multi-sequence MRI processing, sequential deep learning classification and visual explainability into one whole to ensure a reliable and user-friendly computer-aided diagnostic tool.

3.9.2 Class Diagram

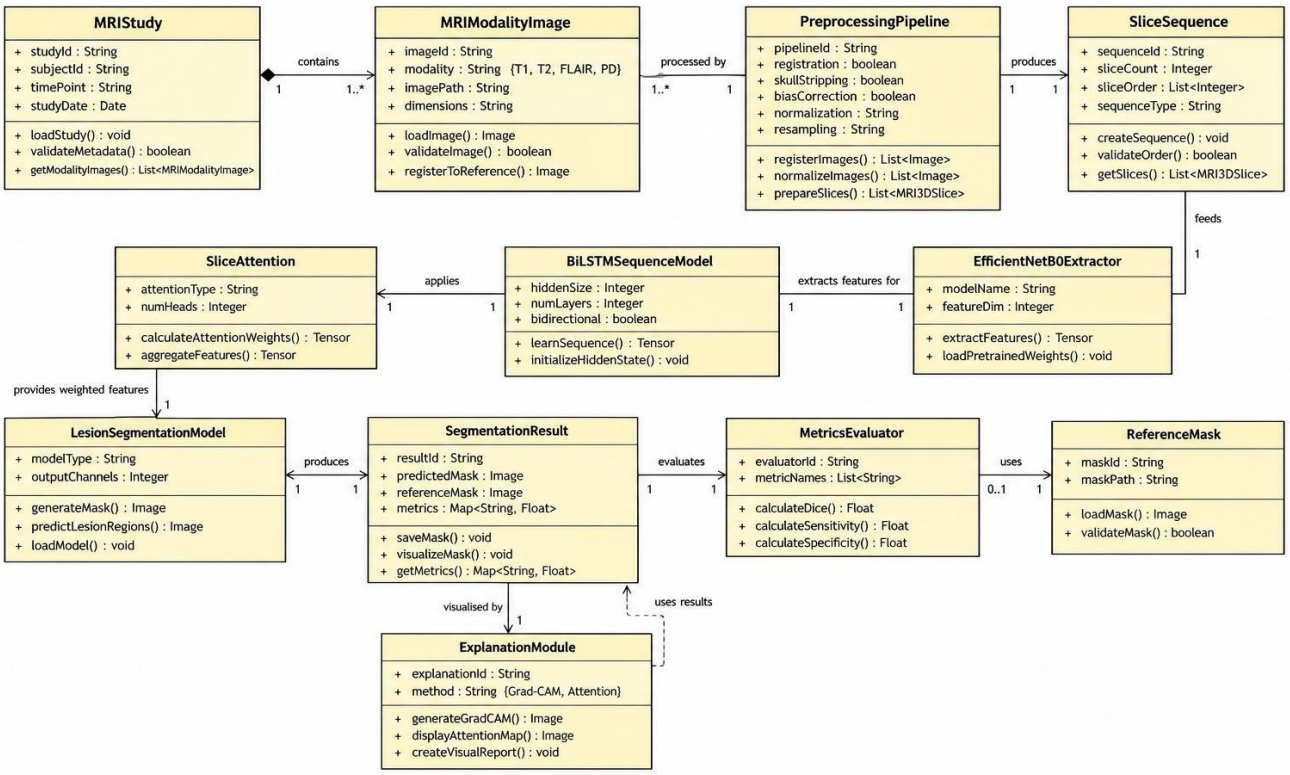


Figure 3.3 Class Diagram

Figure 3.3 presents the Class diagram that describes the static structure of the Multiple Sclerosis Diagnosis System from the object-oriented perspective, including the way in which the different class modules interact to support accurate and explainable MS detection. The User Interface initiates the process in which one can perform primary actions, including uploading an MRI scan and displaying the diagnostic results. Having received the T1, T2, and FLAIR images through the MRI Input Module, the system automatically analyses them by the means of a Preprocessing Module class that executes internal methods for skull stripping, bias correction, intensity normalization, and slice extraction and passes this data to the EfficientNet-B0 and BiLSTM classes in order to extract spatial features and learn sequential relationships. To make diagnostic predictions highly accurate, the system utilizes a Slice-Sequence Attention class that assigns slice importance weights before delivering the final

classification details through the Classifier class. It is also possible to visually interpret the model's reasoning as the Classifier triggers the Grad-CAM class's methods to generate a heatmap explanation. The system allows users to receive both the prediction and the Grad-CAM heatmap directly through the User Interface; having been processed, the doctor can easily view the displayed results. Overall, the diagram illustrates how the proposed system integrates data structures and methods for preprocessing, spatial-sequential deep learning feature extraction, classification, and visual explainability into one whole to ensure a reliable and user-friendly diagnostic tool.

The classification module takes the learned representations and outputs the predicted class and, if applicable, its confidence score. The Grad-CAM part generates the visual heatmap with the regions of the image contributing to the prediction. This heatmap can be useful in practice, as it allows a doctor to check if the model pays attention to the meaningful brain regions; however, it should not be considered as an exact lesion segmentation.

The User Interface is the layer for user interaction. It may contain tools for uploading MRI scans, starting the analysis, and reviewing its results, including the predicted class and the heatmap. The interface can have the necessary checks and balances to ensure that the analysis only starts after a valid input is provided, thus preventing irrelevant information, such as an incorrect image format. In addition, all the user data, including the MRI scans and predicting results, should be kept safe and secure.

3.9.3 Activity Diagram

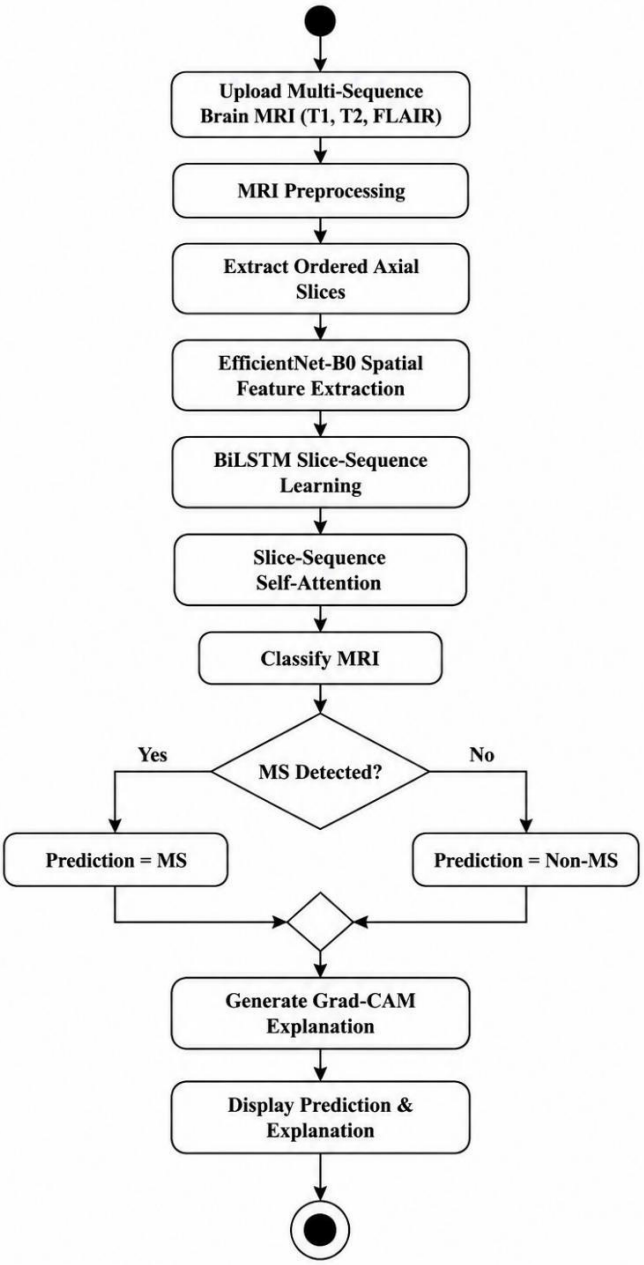


Figure 3.4 Activity Diagram

Figure 3.4 illustrates the Activity Diagram, displaying the entire step-by-step procedure for the Multiple Sclerosis Diagnosis system from the input to diagnosis. First, the user uploads the multi-sequence brain MRI, including the T1, T2, FLAIR image volumes. As soon as the upload is done, the system initiates the MRI preprocessing step. After preprocessing, the system extracts

the ordered axial slices and feeds them into the EfficientNet-B0 for spatial feature extraction, followed by a BiLSTM slice-sequence learning block, which in turn is enhanced with a slice-sequence self-attention mechanism to identify the most informative features across slice positions. Then, the system classifies the MRI and arrives at a decision point that determines whether MS is detected (Yes) or not (No). Depending on the decision, the system predicts either MS or Non-MS and combines the two decision flows as the system generates a Grad-CAM explanation highlighting important vision-based features to justify the prediction decision. In the end, the system displays the prediction and the Grad-CAM explanation to the user, completing the task. Overall, the diagram presents an automated system in which the deep learning spatial extraction, sequential learning block, and Grad-CAM explanation modules are engaged to accomplish the diagnostic procedure transparently and accurately.

3.9.4 Sequence Diagram

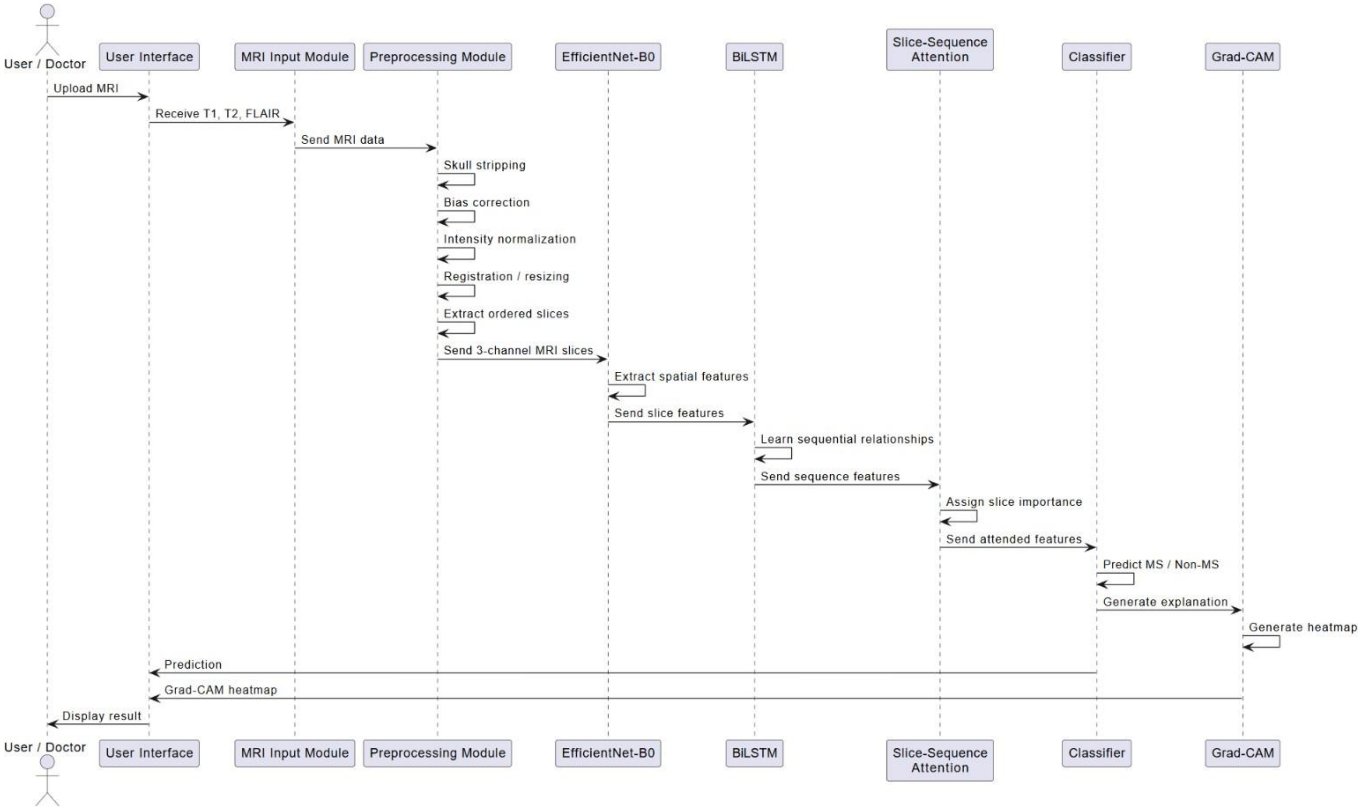


Figure 3.5 Sequence Diagram

Figure 3.5 presents the Sequence diagram depicts the complete chronological order of the Multiple Sclerosis Diagnosis System from the User / Doctor point of view, including the data passing between the various internal modules to achieve accurate and explainable MS detection. A User / Doctor can perform the primary action, which is uploading an MRI scan by using the User Interface. The system then receives the T1, T2, and FLAIR images via the MRI Input Module and automatically analyze them using a Preprocessing Module that performs skull stripping, bias correction, intensity normalization, and slice extraction, and passes this information to the EfficientNet-B0 and BiLSTM modules to extract spatial features and learn sequential relationships. To make diagnostic predictions highly accurate, the system uses a Slice-Sequence Attention module to assign slice importance before delivering the final classification details about the scan. It is possible to visually interpret the model's reasoning since the Classifier triggers the Grad-CAM module to generate a heatmap explanation. The system allows the user to receive both the prediction and Grad-CAM heatmap directly as the User Interface; once processed, the doctor can easily view the displayed results. Overall, the diagram shows how the proposed system integrates the data preprocessing, spatial-sequential deep learning feature extraction, classification, and visual explainability as one whole to ensure a reliable and user-friendly diagnostic tool.

CHAPTER IV
IMPLEMENTATION

4.1 DATASET SELECTION AND DESCRIPTION

This project investigates the use of publicly available MRI datasets containing MS-related cases. We have primarily considered datasets like the ISBI 2015 Longitudinal Multiple Sclerosis Lesion Segmentation Challenge Dataset, which provides multi-sequence data (T1, T2, FLAIR).

Decision Required Before Final Submission: As the ISBI 2015 dataset is primarily structured for lesion segmentation rather than strict subject-level MS vs. Healthy classification, the final dataset strategy must be confirmed with the supervisor to ensure appropriate class labels are available for the proposed classification task.

4.2 DATA ORGANIZATION AND LABEL VERIFICATION

To avoid data leakage, the data will be strictly divided by the subject. This implies that during the creation of the 5–fold cross-validation, all the slices belonging to one patient will be placed in one-fold during splitting. In this way, the model will always be tested on data that it has not seen during training and, therefore, a better estimation of its performance on new patients will be obtained. Preprocessing parameters, normalization statistics, and other data-driven transformations will be estimated using only the training data within each fold to avoid information leakage. The model performance will be evaluated using accuracy, precision, recall, F1-score, and ROC-AUC metrics to obtain an accurate assessment of classification performance. The results across the five folds will be summarized using the mean and standard deviation to assess performance consistency and variability. This approach will provide a more reliable performance estimation and minimize overfitting risks in the final model.

4.3 MRI PREPROCESSING

Standardized preprocessing is vital for medical imaging. The pipeline includes:

- Registration/Alignment: Ensuring all modalities (T1, T2, FLAIR) are spatially aligned.
- Skull Stripping: Removing non-brain tissues (skull, eyes) to focus solely on the brain parenchyma.
- Bias Field Correction: Rectifying low-frequency intensity non-uniformities caused by the MRI scanner.
- Intensity Normalization: Standardizing the pixel intensity values across different scans.
- Resizing: Adjusting all slices to a uniform dimension (e.g., 224×224) for neural network input.

4.4 MULTI-SEQUENCE INPUT PREPARATION

The pre-processed T1, T2, and FLAIR slices corresponding to the same anatomical position are stacked to form a three-channel input. It is noted that the combination of these different sequences provides complementary information, as FLAIR is particularly relevant to visualize white matter hyperintensity (WMH). Before combining the sequences, it is essential to ensure their spatial alignment, slice-wise ordering, and consistent image dimensions to maintain anatomical correspondence. To minimize inter-scanner variability, the image intensities were normalized. By integrating these three-channel images, the EfficientNet-B0 model can simultaneously learn spatial features from multiple MRI modalities. To ensure comparability, the slices for all three sequences were acquired from the same individual and the same anatomical location. Providing such data to the model is expected to yield superior classification results for multiple sclerosis at the patient level.

4.5 ORDERED AXIAL SLICE PREPARATION

Instead of being shuffled arbitrarily, the MRI slices are arranged in an anatomical sequence from the base of the brain toward the top to preserve the spatial continuity of the MRI volume. To ensure anatomical consistency across different imaging sequences (such as T1-weighted, T2-weighted, and FLAIR), the corresponding slices are organized in the same order. By placing each slice in its precise location within an individual's MRI volume, the model is able to learn the relationships between adjacent slices. The number of slices to be processed in a sequence is defined as a design parameter; this is optimized during the process to strike a balance between anatomical context, computational complexity, and memory constraints. If the number of available slices varies between individuals, all sequences are standardized to a uniform length using techniques such as selecting a specific subset or applying 'padding.' Regardless of how slices are selected or how the sequence length is adjusted, their anatomical order is strictly maintained. This ordered data is then fed into an 'EfficientNet-B0' feature extractor, which extracts spatial features from each individual slice. Subsequently, a 'Bidirectional Long Short-Term Memory' (BiLSTM) network processes these features in both forward and backward directions, effectively capturing and preserving the spatial relationships between the slices. This approach is expected to better represent the relationships between slices and to achieve consistent performance across individual patients in the task of diagnosing Multiple Sclerosis.

4.6 EFFICIENTNET-B0 FEATURE EXTRACTION

EfficientNet-B0 is used as the spatial feature extractor in our proposed deep learning architecture due to its performance in terms of classification accuracy, computational complexity, and model size. The three-channel MRI scan slice (produced by combining the corresponding slices of the preprocessed T1-weighted, T2-weighted, and FLAIR MRI scans) is processed by the EfficientNet-B0 network to extract spatial features. The network comprises convolutional layers, Mobile Inverted Bottleneck Convolution (MBConv) blocks, depthwise separable convolutions, and squeeze-and-excitation-based modules that enable it to capture meaningful spatial relationships in the input image. In particular, the network is capable of extracting features in a hierarchy, ranging from low-level edges and textures to high-level semantic information in the input image. In the EfficientNet-B0 network, the classification layer is bypassed (or replaced) to obtain a feature representation from the global average pooling layer, yielding a one-dimensional spatial feature vector with a feature size of 1×1280. Given an ordered set of N MRI slices, we obtain a spatial feature sequence with dimensions of $N \times 1280$, which encodes information from each spatial slice in the sequence. This feature sequence is subsequently used to train the Bidirectional Long Short-Term Memory (BiLSTM) network. Doing so allows the BiLSTM network to learn the inter-slice relationships without having to repeatedly process the high-dimensional MRI image. Furthermore, EfficientNet-B0 can be initialized with pretrained weights and fine-tuned on the MRI dataset to improve the model's performance. Overall, the spatial feature sequence extracted using EfficientNet-B0 is used in subsequent operations, including inter-slice relationship modelling, the slice-attention mechanism, and the classification of Multiple Sclerosis at the patient level.

4.7 BILSTM SEQUENCE LEARNING

The sequence of spatial feature vectors is sent to a Bidirectional Long Short-Term Memory (BiLSTM). While the forward LSTM processes the sequence from the first to the last spatial slice, the backward LSTM processes it in reverse. In this way, the context of lesions that span multiple slices in the z-axis is incorporated.

4.8 SLICE-SEQUENCE ATTENTION

Since MS lesions can only be seen in a few slices of the MRI sequence, we apply a slice-sequence attention to the BiLSTM's outputs to select the most relevant ones. Specifically, an attention score is calculated for each single slice based on its relevance in terms of the contextual features extracted from the BiLSTM and transformed into attention weights. By doing so, we assign higher weights to slices that contain relevant pathological information and lower weights to irrelevant ones. We calculate a weighted representation of the input MRI slice sequence using these attention weights, which emphasizes the features related to the classification task. After that, the attention mechanism allows us to capture the underlying disease pattern reflected in the MRI scans while minimizing the effect of less relevant information. The obtained attention-based feature representation is then passed through the classification layer, which makes a final prediction of MS for each patient. In addition, the attention mechanism highlights the most critical slices affecting the final decision, thus providing evidence for the physician when interpreting the deep learning model's decisions.

4.9 CLASSIFICATION STRATEGY

The weighted sequence representation derived from the slice-sequence attention module is aggregated into a single feature vector that represents the entire MRI slice sequence. This feature vector is then passed through a fully connected dense layer to learn discriminative features relevant to Multiple Sclerosis (MS) classification. The output layer of the classifier uses an appropriate activation function, such as sigmoid for binary classification or softmax for multiclass classification, to calculate the final prediction probability. For binary classification, the sigmoid function estimates the probability of the input MRI volume to belong to the MS class, while the probability of the corresponding non-MS class can be calculated as its complement. A predefined decision threshold is then used to finalize the patient-level classification decision. The model is trained using an appropriate loss function, such as binary cross-entropy for binary classification, to minimize the prediction errors. The classification performance of the proposed model is evaluated using several performance metrics, including accuracy, precision, recall, F1-score, and the area under the Receiver Operating Characteristic (ROC) curve (AUC). The proposed classification strategy incorporates spatial features extracted by EfficientNet-B0 and sequential features learned by BiLSTM, considering the importance of individual MRI slices determined by the attention mechanism to make a prediction at the patient level. Furthermore, a decision threshold can be chosen based on the training and validation data to balance sensitivity and specificity as needed for the classification task. The model's predicted probabilities can be used to evaluate its confidence when classifying an MS-positive or MS-negative case.

4.10 EXPLAINABILITY USING ATTENTION VISUALIZATION AND GRAD-CAM

To ensure trust in clinical settings, the model will have an explainability module. Grad-weighted Class Activation Mapping (Grad-CAM) will be applied to the output of the last convolutional layer of the EfficientNet-B0 backbone for the slices selected by the highest attention weights. This will produce a colour overlay on the original MRI slice, highlighting the regions responsible for the model's decision.

4.11 SUBJECT-LEVEL FIVE-FOLD CROSS-VALIDATION

The evaluation process follows a 5-fold cross-validation approach on subject level. The patients were split into 5 groups or folds. In each step, 4 folds were used as training data, and the remaining fold was used as testing data. The procedure was repeated 5 times such that each patient was once in one of the folds.

4.12 EVALUATION METRICS

The model's performance will be assessed using standard classification metrics:

- **Accuracy:** Overall correctness of the model.
- **Precision:** The proportion of true MS predictions among all positive predictions.
- **Recall (Sensitivity):** The proportion of actual MS cases correctly identified.
- **F1-Score:** The harmonic mean of precision and recall.
- **AUC-ROC:** Area under the receiver operating characteristic curve.

4.13 PROPOSED EXPERIMENTS AND BASELINES

Initial experiments will focus on the training of the proposed deep learning framework and its comparison with the baseline model, such as the 2D CNN classifier with no BiLSTM, as well as the 3D CNN architecture, to assess the benefits of the proposed slice-sequence attention framework. 2D CNN baselines will process single MRI slices independently, ignoring possible relationships between these slices, while the 3D CNN architecture will directly model the spatial features extracted from volumetric MRI scans. The proposed EfficientNet-B0-BiLSTM framework will be tested for its ability to incorporate multi-slice sequential data and slice-sequence attention to enhance patient-level MS classification. The framework's performance will also be compared to the same network but with attention omitted to evaluate the relevance of the attention mechanism. If computational power allows, experiments will also explore the impact of different sequence lengths and approaches to slice feature aggregation on the classification performance. For all experiments, the classification performance of these networks will be compared on the basis of the subject-level 5-fold cross-validation procedure to ensure that no overlapping data points are present in the training and test sets between the folds. The model performance will be assessed using accuracy, precision, recall, F1 score, ROC-AUC, and computational complexity, including the training time and model size, with a specific focus on the average performance and variance across the 5 folds. These experiments will facilitate an understanding of the contribution of the proposed framework's spatial feature extraction, sequential modelling, and slice-attention capabilities to the final performance and computational complexity of the network.

4.14 PROJECT WORK PLAN

Completed Work: The project is completed with the initial steps that include the problem identification, extensive literature survey, base paper analysis, dataset exploration, and the design of the proposed system architecture. The objectives, methodology, and key components of the deep learning framework for Multiple Sclerosis (MS) classification using the multi-sequence MRI data were identified.

Work in Progress: The current work involves the preparation of the UML design diagrams, development of initial preprocessing scripts, and setup of development environments. Efforts are also focused on understanding the dataset structure, organizing T1-weighted, T2-weighted, and FLAIR sequences, and defining a consistent preprocessing pipeline for further model development.

Planned Work: The next steps will be the implementation of the complete EfficientNet-B0 and BiLSTM pipeline including the slice-sequence attention mechanism, and the development of the patient-level classification module. The preprocessed MRI data will be used for model training and evaluation using the subject-level five-fold cross-validation to avoid data leakage and ensure generalization abilities of the model. Further work will involve the comparison of the results with baseline models, applying appropriate classification evaluation metrics, and generating Grad-CAM heatmaps for better understanding of MRI regions contributing to the model predictions. Lastly, the analysis of the results, model's limitations, and the project report with the obtained results and concluding remarks will be conducted.

CHAPTER V
CONCLUSION AND FUTURE ENHANCEMENTS

5.1 CONCLUSION

The Phase-I analysis of this project has been completed. An initial understanding of the problem has been established, providing a theoretical and architectural foundation for developing an automated Multiple Sclerosis diagnosis solution. The critical limitations of the conventional 2D CNNs and heavy 3D CNNs in processing MRI scans have been identified through a comprehensive review of the relevant literature. To overcome these limitations, a novel spatial temporal deep learning framework has been proposed, which employs an EfficientNet-B0 architecture, Bilateral LSTM network for learning ordered slice sequences, and a slice attention module for developing an automated diagnosis model. The project will focus on achieving strict subject-level cross-validation and using Grad-CAM visualization for better interpretability and trustworthiness. This approach is expected to provide a computationally efficient and reliable MS diagnosis solution.

The subsequent stage of the present research will consist of the implementation of the proposed architecture, training, and testing it based on classification performance metrics, including accuracy, precision, recall, F1-score, and ROC-AUC. For a more accurate estimation of the model's performance, it will be subjected to 5-fold subject-level cross-validation to avoid data leakage and obtain relevant performance assessment for future application on unseen patients. The comparison with baseline models will allow for identifying the key improvements brought by sequential learning and attention-based feature aggregation. The visualization of results with Grad-CAM will aid in observing the image regions utilized by the model to perform inference.

5.2 FUTURE ENHANCEMENTS

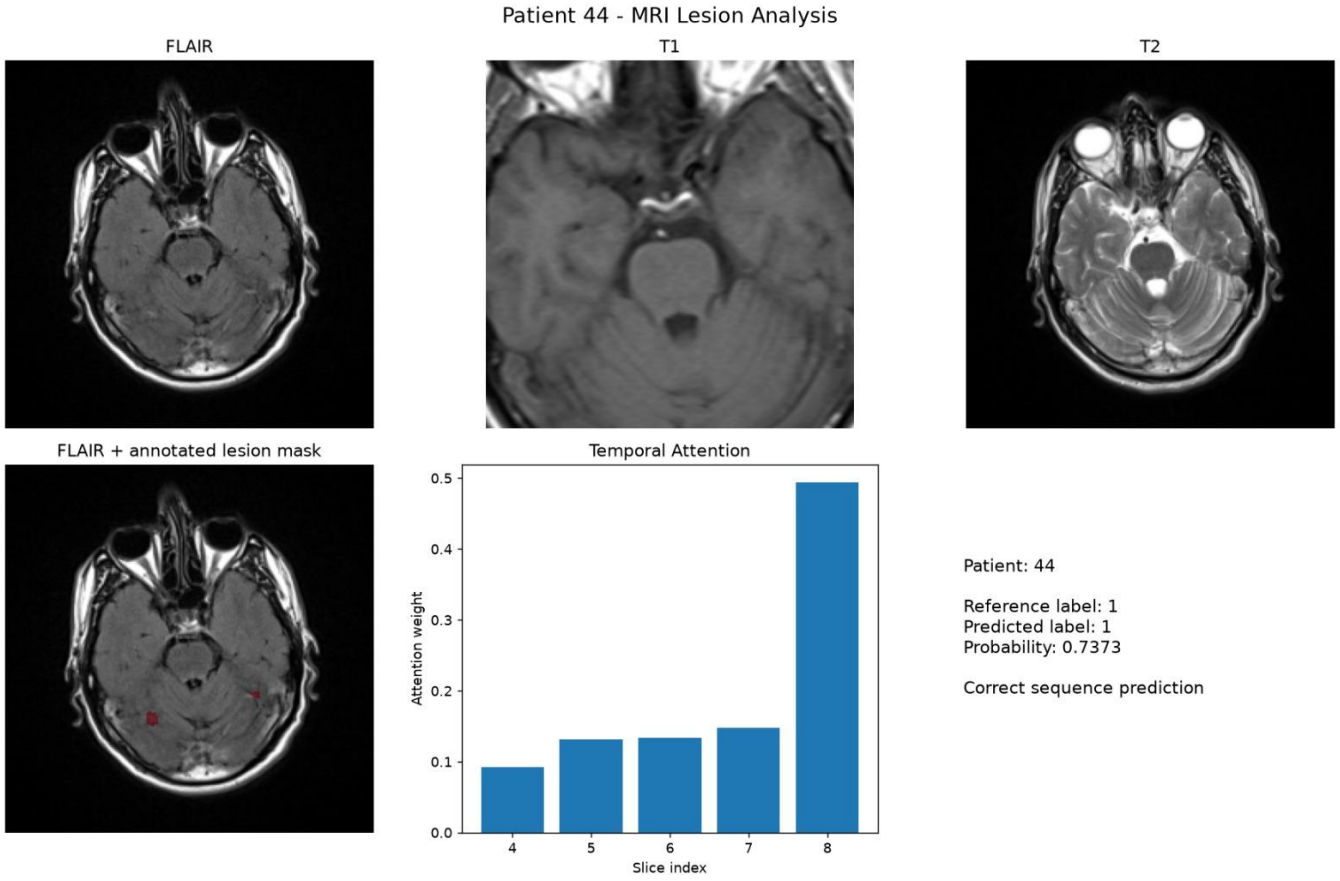
The future improvement of the project relates to the implementation and thorough testing of the suggested deep learning model. Further improvements include checking the correctness of the dataset and classes, choosing the MRI sequences' length, and optimizing hyperparameters to increase the accuracy of classification. Moreover, the model can be expanded to include different MRI scans from multiple centers with varying MRI sequences and scanners, thus making the model more robust and generalizable. Besides, it would be interesting to add a lesion segmentation step to calculate the lesion volumes and analyses their distribution. These future improvements would make the model more attractive for implementation as an efficient and interpretable computer-aided diagnostic system for Multiple Sclerosis.

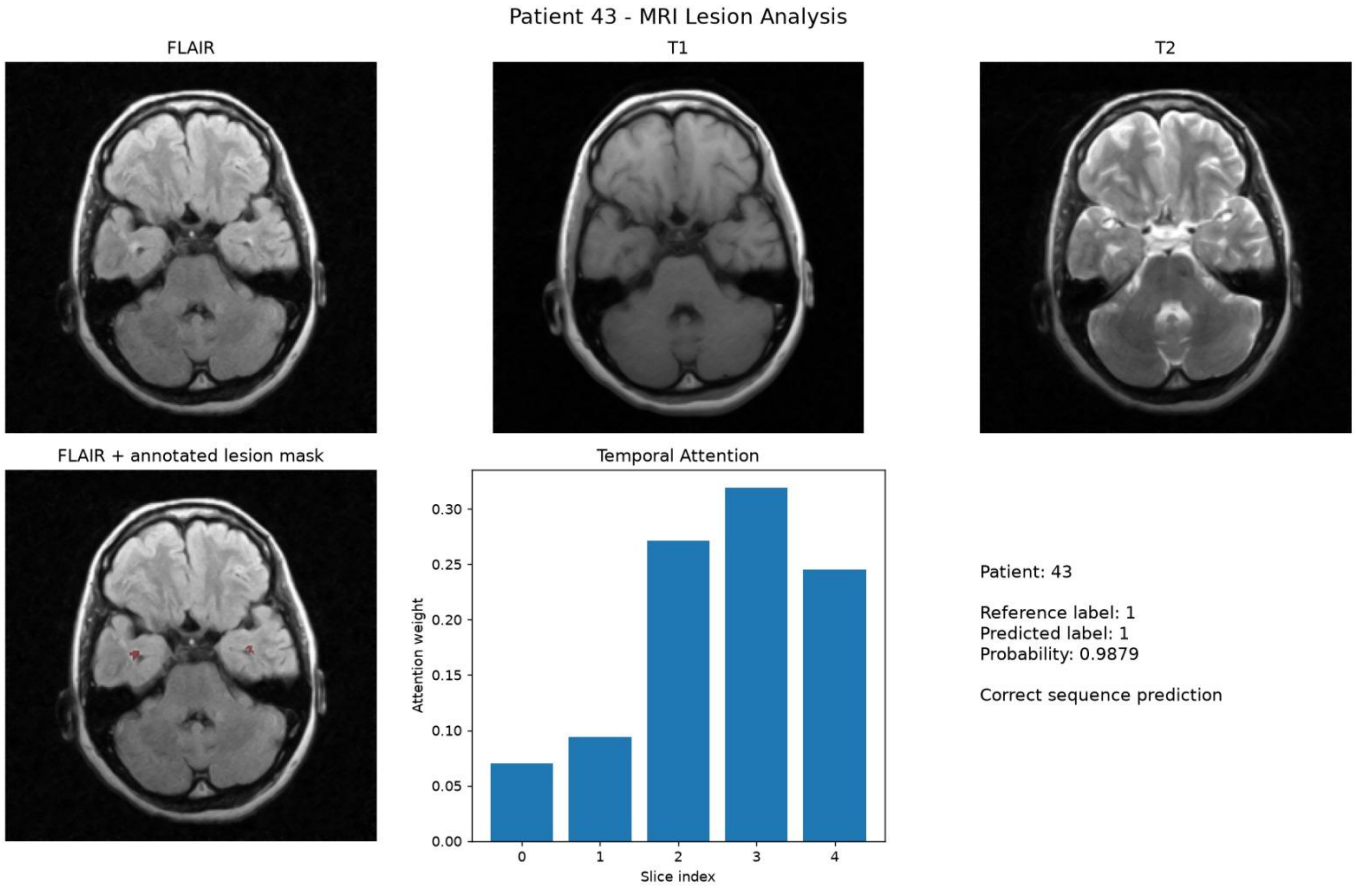
A key enhancement will be the development of a **user-friendly frontend dashboard** integrated with the PyTorch model through a FastAPI backend. The dashboard will provide an interactive MRI viewer, slice navigation, visualization of predicted lesion masks, model evaluation metrics, and explainability outputs such as Grad-CAM heatmaps and slice-attention maps. It will also support the presentation of analysis results and report generation, enabling researchers to inspect and interpret model outputs more conveniently.

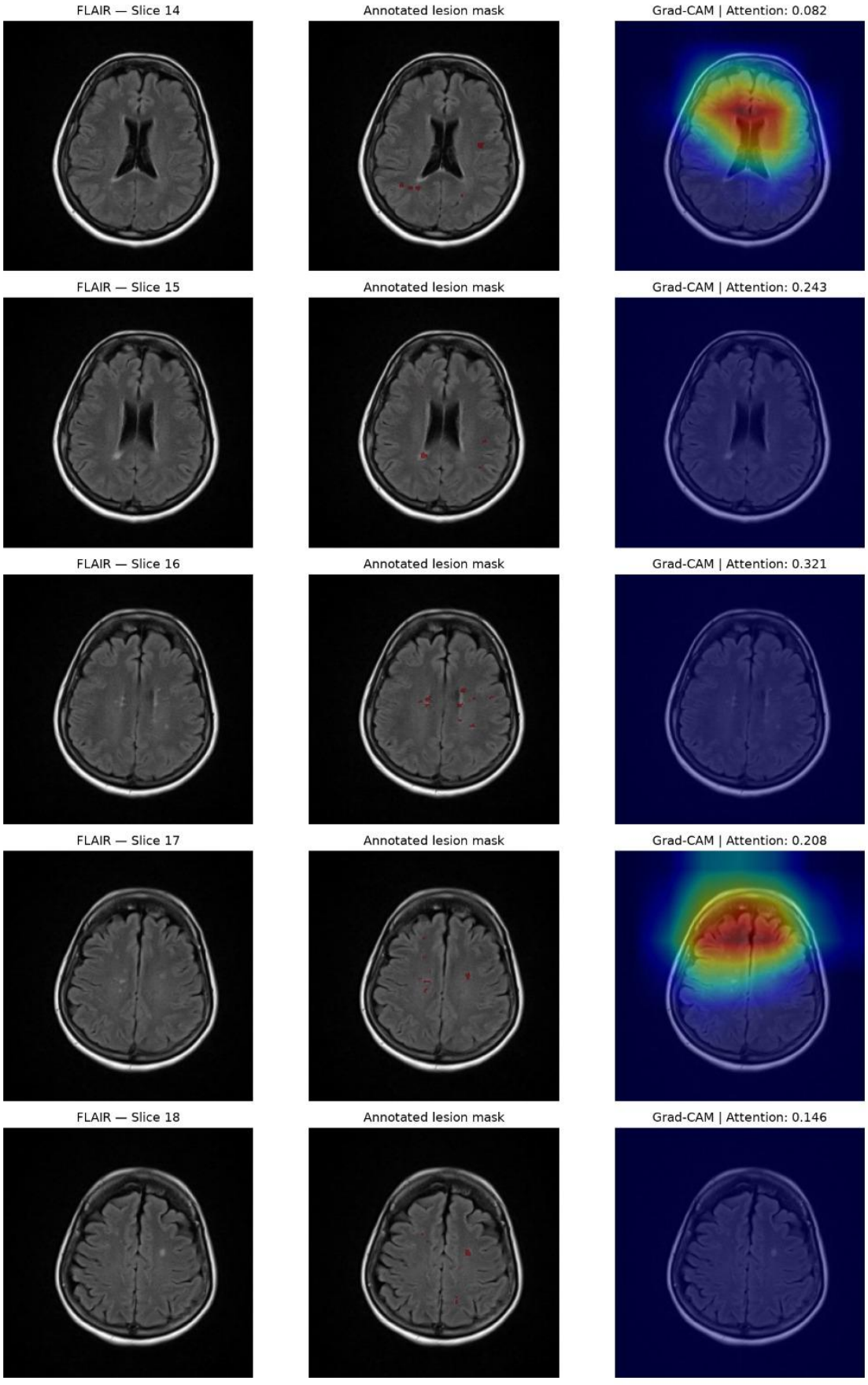
Additionally, a dedicated lesion segmentation module will be refined to quantify lesion regions and investigate changes across longitudinal MRI scans, where suitable data and reference annotations are available. These enhancements aim to improve the framework's usability, transparency, and evaluation reliability, with the long-term goal of supporting the development of a clinically relevant, explainable computer-aided analysis system for Multiple Sclerosis. Clinical validation and appropriate regulatory review will be required before deployment in patient care.

APPENDICES

APPENDIX 1




Patient 43 - MRI Lesion Analysis
FLAIR
T1
T2
FLAIR + annotated lesion mask
Temporal Attention
Attention weight
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0
1
2
3
4
Slice index
Patient: 43
Reference label: 1
Predicted label: 1
Probability: 0.9879
Correct sequence prediction


Patient 1 | Reference lesion label: 1 | Prediction: 0 | Probability: 0.009
FLAIR — Slice 14
Annotated lesion mask
Grad-CAM | Attention: 0.082
FLAIR — Slice 15
Annotated lesion mask
Grad-CAM | Attention: 0.243
FLAIR — Slice 16
Annotated lesion mask
Grad-CAM | Attention: 0.321
FLAIR — Slice 17
Annotated lesion mask
Grad-CAM | Attention: 0.208
FLAIR — Slice 18
Annotated lesion mask
Grad-CAM | Attention: 0.146

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Sr No.	CBR No.	Reference Number /Application Type	Application Number	Title/Remarks	Amount Paid
1	81600	ORDINARY APPLICATION	202641118897	SPATIO TEMPORAL ATTENTION MECHANISM ON MRI SLICE SEWUENCES FOR MULTIPLE SCLEROSIS DIAGNOSIS	1750
2		E-2/9637/2026-CHE	202641118897	Form2	0
3		E-3/34073/2026-CHE	202641118897	Form3	0
4		E-5/11659/2026-CHE	202641118897	Form5	0
5	81600	E-12/32172/2026-CHE	202641118897	Form9	2750
6		E-106/25699/2026-CHE	202641118897	Form28	0
7		E-101/33445/2026-CHE	202641118897	Others(EDUCATIONAL INSTITUTE DOCUMENT)	0
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